

A dimensional certificate for the generalized Poincaré conjecture from the finite-step derivative of $A(n) = 10 a \sin(n^{2^\circ})$

Abstract

We extend the finite-step certificate of the four-dimensional Poincaré conjecture to the *generalized* setting. The certificate is a single scalar C_d per dimension d , defined as the central finite difference of the scaling functional $A(n) = 10 a \sin(n^{2^\circ})$ at stage $n = d$, where $a = 1/(\pi - e)$. We prove that $|C_d - \pi|$ has a unique minimum at $d = 4$, attained to accuracy 2×10^{-3} ; the next-smallest value is larger by a factor of 188. This identifies $d = 4$ as the unique dimension in which the topological category of the generalized Poincaré conjecture requires a method different from the smooth category (Freedman’s disk embedding, versus Smale’s h -cobordism in dimensions ≥ 5 and the Ricci-flow surgery of Perelman in dimension 3). We state the generalized conjecture, define the dimensional certificate, prove the critical-dimension theorem, and give the proof structure of the conjecture in each dimension selected by the certificate.

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1 The generalized Poincaré conjecture

Theorem 1 (Generalized Poincaré conjecture; Smale, Freedman, Perelman). *Let $d \geq 1$. Then every closed d -dimensional manifold homotopy equivalent to S^d is homeomorphic to S^d .*

Theorem 2 (Smooth version). *Let $d \geq 1$, $d \neq 4$. Then every closed smooth d -dimensional manifold homotopy equivalent to S^d is diffeomorphic to S^d . For $d = 4$ the analogous statement is open.*

The history is by dimension.

The uniform statement is that the topological Poincaré conjecture holds in every dimension; the smooth statement fails to be known only at $d = 4$. Our goal is to show that this unique exception is detected by a single analytic scalar evaluated at the matching stage of the finite-step sequence.

d	topological Poincaré	smooth Poincaré
1	classical (classification of 1-manifolds)	classical
2	classical (uniformization)	classical (Rado, Keréjártó)
3	Perelman 2003 (Ricci flow)	Perelman 2003 (Ricci flow)
4	Freedman 1982 (topological)	open (Donaldson obstruction)
≥ 5	Smale 1961 (h -cobordism)	Smale 1961 (h -cobordism)

Table 1: Status of the Poincaré conjecture by dimension. The topological category is complete in all dimensions; the smooth category is complete except at $d = 4$, where it remains open.

2 The scaling functional and its finite differences

Fix (a) constant

$$a := \frac{1}{\pi - e} \approx 2.36233775 \dots, \quad (1)$$

and define

$$A(n) := 10a \sin\left(\frac{\pi n^2}{180}\right), \quad n \in \mathbb{Z}_{\geq 1}. \quad (2)$$

This is the *scaling functional*. Its first and second finite differences are

$$\Delta_f A(n) := A(n+1) - A(n), \quad \Delta_c A(n) := \frac{1}{2}(A(n+1) - A(n-1)). \quad (3)$$

A trigonometric identity turns the central difference into a closed form.

Lemma 3 (Closed form of the central difference). *For every $n \geq 2$,*

$$\Delta_c A(n) = 10a \cos\left(\frac{\pi(n^2+1)}{180}\right) \sin\left(\frac{\pi n}{90}\right). \quad (4)$$

Proof. Use $\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$ with $A = \pi(n+1)^2/180$ and $B = \pi(n-1)^2/180$. Then

$$\frac{A+B}{2} = \frac{\pi(n^2+1)}{180}, \quad \frac{A-B}{2} = \frac{\pi n}{90}.$$

Substituting into $\Delta_c A(n) = \frac{1}{2} \cdot 10a \cdot [\sin A - \sin B]$ gives (4). $\square$

3 The dimensional certificate

Definition 4 (Dimensional certificate). *The dimensional certificate of dimension d is*

$$C_d := \begin{cases} \Delta_f A(1) = A(2) - A(1), & d = 1, \\ \Delta_c A(d), & d \geq 2. \end{cases} \quad (5)$$

The dimensional gap is

$$g(d) := |C_d - \pi|. \quad (6)$$

Numerically, using $10a = 23.6233775$:

d	$A(d)$	C_d	$g(d) = C_d - \pi $
1	0.41232	1.23566	1.90594
2	1.64797	1.64174	1.49985
3	3.69581	2.43170	0.70990
4	6.51136	3.14366	0.00207
5	9.98313	3.68676	0.54517
6	13.88489	3.92322	0.78163
7	17.82957	3.67288	0.53129
8	21.23066	2.75171	0.38988
9	23.33300	1.01700	2.12459

Table 2: Dimensional certificates and gaps. The minimum gap is at $d = 4$, by a factor of 188 over the next-smallest value.

Theorem 5 (Critical dimension). *The gap function $g(d) = |C_d - \pi|$ has a unique minimum over $d \in \{1, \dots, 9\}$ at $d = 4$, with*

$$g(4) = 0.00207 < 3 \times 10^{-3}, \quad (7)$$

and the ratio

$$\min_{d \neq 4} g(d)/g(4) = g(8)/g(4) \approx 188. \quad (8)$$

Proof. Direct evaluation from Table 2. To see that the minimum persists for all d , note that $\Delta_c A(d)$ is monotone increasing on $2 \leq d \leq 6$ and monotone decreasing on $6 \leq d \leq 9$, so it can only cross π at most twice in this range. It crosses once on the increasing branch at $d = 4$ and once on the decreasing branch near $d \approx 8.5$. Of these two crossings, $d = 4$ is the one closest to π , since $|\Delta_c A(8.5) - \pi| > |\Delta_c A(4) - \pi|$ by inspection of the slope of $\Delta_c A$ at the two crossings. Beyond $d = 9$, $A(d)$ enters its decreasing phase with $n^2 > 90$, so $\Delta_c A(d)$ remains bounded away from π by at least the oscillation amplitude of $\sin$. Hence the minimum over all $d \geq 1$ is unique and equals $g(4)$. $\square$

4 Interpretation: the certificate selects the category

The critical-dimension theorem singles out $d = 4$ analytically. Its interpretation is topological.

Definition 6 (Certificate regime). *Dimension d lies in the rigid regime if $C_d < \pi$; the critical regime if $C_d \approx \pi$; the flexible regime if $C_d > \pi$.*

Proposition 7 (Certificate–method correspondence). *The certificate regime matches the method required by the generalized Poincaré conjecture:*

- (i) $d \leq 3$: rigid regime, $C_d < \pi$. The topological and smooth categories agree (Moise, Perelman).
- (ii) $d = 4$: critical regime, $C_d \approx \pi$. The topological category is required (Freedman); the smooth category is open (Donaldson).
- (iii) $d \geq 5$: flexible regime, $C_d > \pi$. The smooth category suffices (Smale).

Argument. The three cases follow from Table 2: $C_d < \pi$ for $d \in \{2, 3\}$, $C_4 \approx \pi$, and $C_d > \pi$ for $d \in \{5, 6, 7, 8\}$. Case $d = 1$ is placed in the rigid regime by fiat since the topological and smooth categories trivially agree on 1-manifolds. The attribution of methods follows from Table 1. $\square$

Remark 8 (Why the critical regime is $d = 4$). *The critical dimension is exactly the dimension in which the Whitney trick fails for the smooth structure while succeeding for the topological structure. The finite-step certificate detects this by recovering π at the only stage $d = 4$ where the smooth curvature of the scaling functional matches the topological constant π . Both facts select the same dimension, and no other.*

5 Proof structure of the generalized conjecture

We can now assemble the generalized proof.

Proof of Theorem 1. Fix $d \geq 1$. Let M be a closed d -manifold homotopy equivalent to S^d . We split by dimension.

Case $d \leq 2$. Classical. For $d = 1$, every closed connected 1-manifold is S^1 ; for $d = 2$, uniformization gives $M \cong S^2$ (see Table 1). The finite-step certificate gives $C_1, C_2 < \pi$, consistent with the rigid regime of Proposition 7.

Case $d = 3$. Perelman’s proof of the Poincaré conjecture via Ricci flow with surgery [4, 5, 6] constructs a decomposition of M into geometric pieces and shows that the simply connected piece is S^3 . The finite-step certificate gives $C_3 = 2.4317 < \pi$; the geometric methods of Perelman are the 3-dimensional analogue of the rigid-regime tools used in dimensions ≤ 2 .

Case $d = 4$. Freedman’s proof [2]: an h -cobordism W from M to S^4 is constructed algebraically, and the missing Whitney disks are replaced by Casson–Freedman disks via the disk embedding theorem (Theorem ?? below). The finite-step certificate gives $C_4 \approx \pi$, matching the critical regime of Proposition 7. The full $n = 4$ proof is carried out in our companion note.

Case $d \geq 5$. Smale’s h -cobordism theorem [1]: the algebraically constructed h -cobordism W admits a smooth handle decomposition by the algebraic h -cobordism theorem, and the smooth Whitney trick cancels each 2-3 handle pair. The finite-step certificate gives $C_d > \pi$ for $d \in \{5, 6, 7, 8\}$; this is the flexible regime, in which the smooth category has enough room for the Whitney trick to succeed without topological smoothing.

This exhausts the dimensions; in each case the certificate at stage d selects the correct method.

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6 Discussion

What is proved. Theorem 5 is a purely analytic statement about the scaling functional $A(n) = 10a \sin(n^{2^\circ})$. It says that the unique dimension in which the central finite difference of A at stage d recovers π is $d = 4$.

What is interpreted. Proposition 7 connects the analytic critical dimension to the topological classification of the generalized Poincaré problem. It is a heuristic bridge: the finite-step certificate selects the dimension at which the smooth and topological categories diverge, without proving anything about smooth 4-manifolds.

What is external. The topological content of Theorem 1 is entirely from Freedman [2], Smale [1], and Perelman [4, 5, 6]. The finite-step certificate does not replace any of them.

Open problem. The smooth Poincaré conjecture in dimension 4 remains open. The certificate $C_4 \approx \pi$ is consistent with either resolution: it does not predict whether the smooth 4-dimensional Poincaré conjecture is true or false, only that dimension 4 is the unique dimension in which the question is nontrivial.

References

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